

常见分布 的 数学期望以及方差公式

原创

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4 订阅

4 篇文章

订阅专栏

一、通用公式【数学期望】

1》求解数学期望

1.离散型

$$E(X) = \sum_{k=1}^{\infty} x_k p_k$$

2.连续型

$$E(X) = \int_{-\infty}^{+\infty} x f(x) dx$$

3.Y=g(X)

$$E(Y) = E[g(X)] = \sum_{k=1}^{\infty} g(x_k) p_k$$

$$E(Y) = E[g(X)] = \int_{-\infty}^{+\infty} g(x) f(x) dx$$

Z=g(X,Y)

$$E(Z) = E[g(X,Y)] = \sum_i \sum_j g(x_i, y_j) p_{ij}$$

$$E(Z) = E[g(X,Y)] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x, y) f(x, y) dx dy$$

2》数学期望的性质

性质1 $E(C) = C$ ； 其中C为常数.

性质2 $E(CX) = CE(X)$

性质3 $E(X+Y) = E(X) + E(Y)$

性质4 设X, Y是两个相互独立的随机变量, 则有

$$E(XY) = E(X) \cdot E(Y)$$

推广

(1) $E(X_1+X_2+...+X_n) = E(X_1) + E(X_2) + ... + E(X_n)$

(2) $E(k_1X_1+k_2X_2+...+k_nX_n) = E(k_1X_1)+E(k_2X_2)+...+E(k_nX_n)$
 $= k_1E(X_1)+k_2E(X_2)+ ... + k_nE(X_n)$

(3) 若 $X_1, X_2, ..., X_n$ 相互独立,

则 $E(X_1X_2...X_n) = E(X_1)E(X_2)...E(X_n)$

特别 $E(E(X)) = E(X)$

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二、常用分布的期望与方差

1》精简版：

分布	0 - 1	B (n, p)	$P(\lambda)$	$U(a, b)$	$E(\lambda)$	$N(\mu, \sigma^2)$
$E(X)$	p	np	λ	$(a+b)/2$	$1/\lambda$	μ
$D(X)$	pq	npq	λ	$(b-a)^2/12$	$1/\lambda^2$	σ^2

2》叨叨版：

1) 0-1分布 概率分布为

X	1	0
p_k	p	$1-p$

$$E(X) = 1 \times p + 0 \times (1-p) = p$$

$$D(X) = E(X^2) - [E(X)]^2 = p - p^2 = p(1-p) = pq$$

2) 二项分布 设随机变量 $X \sim B(n, p)$ ，其概率分布为：

$$P\{X = k\} = C_n^k p^k q^{n-k}, \quad k = 0, 1, 2, \dots, n, \quad q = 1 - p$$

$$\begin{aligned} E(X) &= \sum_{k=0}^n k \cdot p_k = \sum_{k=0}^n k \cdot C_n^k p^k q^{n-k} = \sum_{k=0}^n k \cdot \frac{n!}{k!(n-k)!} p^k q^{n-k} \\ &= \sum_{k=1}^n \frac{n!}{(k-1)!(n-k)!} p^k q^{n-k} = np \sum_{k=1}^n \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} q^{n-k} \\ &= np \sum_{k=1}^{n-1} C_{n-1}^{k-1} p^{k-1} q^{(n-1)-(k-1)} = np(p+q)^{n-1} = np. \end{aligned}$$

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2) 二项分布 设随机变量 $X \sim B(n, p)$ ，其概率分布为：

$$P\{X = k\} = C_n^k p^k q^{n-k}, \quad k = 0, 1, 2, \dots, n, \quad q = 1 - p$$

$$D(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = E[X(X-1) + X] = E[X(X-1)] + E(X)$$

$$= \sum_{k=0}^n k(k-1) C_n^k p^k q^{n-k} + np$$

$$= \sum_{k=0}^n k(k-1) \frac{n!}{k!(n-k)!} p^k q^{n-k} + np$$

$$= n(n-1)p^2 \sum_{k=2}^n \frac{(n-2)!}{(k-2)!(n-k)!} p^{k-2} q^{n-k} + np$$

$$= n(n-1)p^2(p+q)^{n-2} + np = n(n-1)p^2 + np$$

$$\text{所以 } D(X) = E(X^2) - [E(X)]^2 = n(n-1)p^2 + np - n^2p^2 = npq$$

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3) 泊松分布 设随机变量 $X \sim P(\lambda)$, 其概率分布为:

$$P\{X = k\} = \frac{\lambda^k}{k!} e^{-\lambda} \quad , \quad k = 0, 1, 2, 3, \dots, \lambda > 0$$

$$\begin{aligned} E(X) &= \sum_{k=0}^{\infty} k \cdot \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} e^{-\lambda} \\ &= \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \\ &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{(j)!} = \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda \end{aligned}$$

(注意: $\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$)

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3) 泊松分布 设随机变量 $X \sim P(\lambda)$, 其概率分布为:

$$P\{X = k\} = \frac{\lambda^k}{k!} e^{-\lambda} \quad , \quad k = 0, 1, 2, 3, \dots, \lambda > 0$$

$$D(X) = E(X^2) - [E(X)]^2$$

$$\begin{aligned} E(X^2) &= E[X(X-1) + X] = E[X(X-1)] + E[X] \\ &= \sum_{k=0}^{\infty} k(k-1) \frac{\lambda^k}{k!} e^{-\lambda} + \lambda \\ &= \lambda^2 e^{-\lambda} \cdot e^{\lambda} + \lambda = \lambda^2 + \lambda \end{aligned}$$

因此, $D(X) = E(X^2) - [E(X)]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$

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4) 均匀分布 设 $X \sim U[a, b]$ 概率密度为:

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{其它} \end{cases}$$

$$E(X) = \int_{-\infty}^{+\infty} xf(x)dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{a+b}{2}$$

$$E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x)dx = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{a^2 + ab + b^2}{3}$$

$$\begin{aligned} D(X) &= E(X^2) - [E(X)]^2 = \frac{1}{3}(a^2 + ab + b^2) - \left(\frac{a+b}{2}\right)^2 \\ &= \frac{(b-a)^2}{12} \end{aligned}$$

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5) 指数分布 设 $X \sim E(\lambda)$ 概率密度为:

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$\begin{aligned} E(X) &= \int_{-\infty}^{+\infty} xf(x)dx = \int_0^{+\infty} \lambda x e^{-\lambda x} dx = -\int_0^{+\infty} x de^{-\lambda x} \\ &= -[xe^{-\lambda x} \Big|_0^{+\infty} - \int_0^{+\infty} e^{-\lambda x} dx] = \int_0^{+\infty} e^{-\lambda x} dx = \frac{1}{\lambda} \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{+\infty} x^2 f(x)dx = \int_0^{+\infty} \lambda x^2 e^{-\lambda x} dx \xrightarrow{\text{令 } \lambda x = t} \frac{1}{\lambda^2} \int_0^{+\infty} t^2 e^{-t} dt \\ &= \frac{1}{\lambda^2} \Gamma(3) = \frac{2}{\lambda^2} \quad \text{故, } D(X) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2} \end{aligned}$$

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6) 正态分布 设 $X \sim N(\mu, \sigma^2)$ 概率密度为:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad (-\infty < x < +\infty)$$

$$E(X) = \int_{-\infty}^{+\infty} x \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

令 $\frac{x-\mu}{\sigma} = t$, 得

$$\begin{aligned} E(X) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} (\sigma t + \mu) e^{-\frac{t^2}{2}} dt = \mu \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt \\ &= \mu \cdot \end{aligned}$$

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6) 正态分布 设 $X \sim N(\mu, \sigma^2)$ 概率密度为:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad (-\infty < x < +\infty)$$

$$\begin{aligned} D(X) &= E[X - E(X)]^2 = \int_{-\infty}^{+\infty} (x - \mu)^2 \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\ &= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} t^2 e^{-\frac{t^2}{2}} dt \quad (t = \frac{x-\mu}{\sigma}) \\ &= \frac{\sigma^2}{\sqrt{2\pi}} \left[t(-e^{-\frac{t^2}{2}}) \Big|_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} e^{-\frac{t^2}{2}} dt \right] \\ &= \frac{\sigma^2}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = \sigma^2 \end{aligned}$$

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